

- 3 Stomata in leaves respond to changes in environmental conditions by opening and closing. This regulates carbon dioxide uptake and water loss.

(a) Describe the mechanism by which stomata open in sunlight.

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- (b) Stomatal conductance is a measure of the rate of water vapour loss from the intercellular air spaces of leaves to the atmosphere through the stomata.

A student investigated the effect of different concentrations of abscisic acid (ABA) on stomatal conductance in *Helianthus annuus*, the common sunflower.

The student treated the roots of 15 sunflower plants with different concentrations of ABA solution. The plants were left in standardised conditions for 24 hours.

After 24 hours, the student measured:

- the concentration of ABA in the xylem sap of each plant
- the stomatal conductance of the leaves of each plant.

The results are shown in Fig. 3.1.

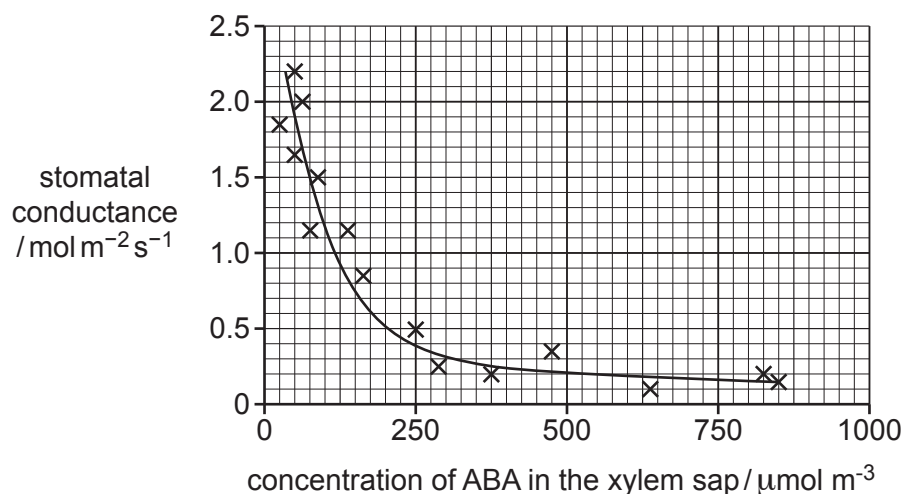


Fig. 3.1

- (i) With reference to Fig. 3.1, describe the relationship between the concentration of ABA in the xylem sap and stomatal conductance.

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- (ii) Suggest explanations for the results shown in Fig. 3.1.

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- (c) Some transcription factors increase the rate of transcription of genes involved in the closure of stomata.

State where transcription factors bind to cause an increase in the rate of transcription.

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[Total: 11]